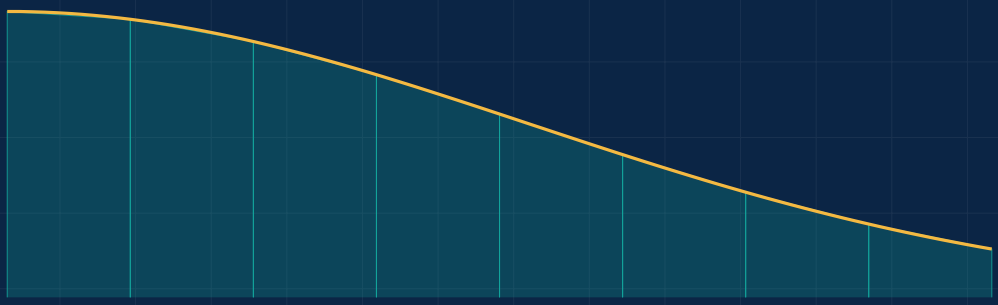
Numerical Integration

A Short Course

Numerical Integration

A Short Course

From Riemann Sums to Gaussian Quadrature



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This book is a typesetting sample. It was produced entirely with LaTeX (pdfLaTeX, `book` class, TikZ, `pgfplots`, `biblatex` and `makeindex`). All tables and numerical results are computed by the script `book_data.py` distributed with the source files.

First edition, October 2026.

To the students who ask why the area under a curve
cannot always be computed exactly,
and then go on to compute it anyway.

Preface

Integrals are everywhere, but exact antiderivatives are rare. This short book is a self-contained introduction to the numerical evaluation of definite integrals in one dimension. It starts from the simplest idea, a sum of areas of rectangles, and ends with two of the most useful tools of numerical analysis: Gaussian quadrature and adaptive quadrature.

The book follows one running example, the integral

$$I = \int_0^1 e^{-x^2}\,\mathrm{d}x \approx 0.7468241328,$$

which has no elementary antiderivative but is perfectly suited to experiments. The experiments are reproducible: every table and data file is generated by a short Python script, and the numbers quoted in the text are taken from its output.

How to read this book. Chapter 1 introduces quadrature rules and the notions of degree of exactness and order of convergence. Chapter 2 studies the Newton–Cotes rules, with error estimates and extrapolation. Chapter 3 presents Gaussian quadrature. Chapter 4 deals with practical questions: rounding errors, adaptive algorithms and the choice of a method. Exercises are collected at the end of each chapter, and solutions to all of them are given in Appendix B.

A note on this edition. This is also a demonstration of professional book typesetting with LaTeX: running headers, theorem boxes, numbered exercises, cross-references, a bibliography and an index are all generated automatically.

N. S. Y.

Natitingou, October 2026

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1

Why Integrate Numerically?

1.1 Integrals without closed forms

The fundamental theorem of calculus tells us to compute $\int_a^b f(x)\,\mathrm{d}x$ by finding an antiderivative F and evaluating $F(b) - F(a)$. This works for polynomials, for $\sin$, $\exp$ and many more functions, but it fails far more often than introductory courses suggest. By a theorem of Liouville, the function e^{-x^2} has no antiderivative that can be written with elementary functions. The same is true of $\sin(x)/x$ and of the integrands that define elliptic integrals.

One can give the missing antiderivative a name: the *error function*

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t.$$

But this only moves the problem: to obtain a number such as $\operatorname{erf}(1)$, one still has to approximate an integral. Numerical integration, also called *quadrature*, is therefore a basic tool of scientific computing. It is also needed when f is known only through measurements, that is, through a table of values.

Example 1.1: The running example

Throughout this book we use

$$I = \int_0^1 e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}\operatorname{erf}(1) \approx 0.7468241328.$$

The exact value is known to many digits, so that we can measure the error of every method.

1.2 Quadrature rules and degree of exactness

Definition 1.2: Quadrature rule

A *quadrature rule* with m *nodes* $x_1, \dots, x_m \in [a, b]$ and *weights* $w_1, \dots, w_m$ is the approximation

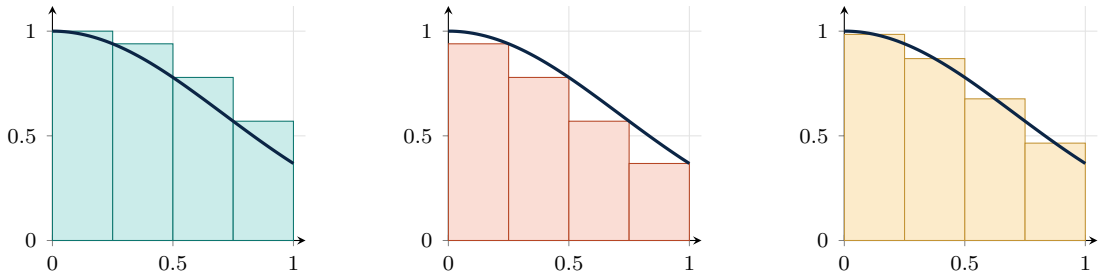
$$Q(f) = \sum_{i=1}^{m} w_i\, f(x_i) \approx \int_a^b f(x)\,\mathrm{d}x.$$

Its *degree of exactness* is the largest integer d such that $Q(p) = \int_a^b p$ for every polynomial p of degree at most d.

The simplest rules are the Riemann sums. Let $h = (b-a)/n$ and $x_i = a + ih$. The *left*, *right* and *midpoint* sums are

$$L_n = h\sum_{i=0}^{n-1} f(x_i), \qquad R_n = h\sum_{i=1}^{n} f(x_i), \qquad M_n = h\sum_{i=0}^{n-1} f\left(x_i + \tfrac{h}{2}\right). \tag{1.1}$$

Figure 1.1 shows the three rules for $n = 4$ on the running example. The left and right sums have degree of exactness 0 (they are exact for constants only); the midpoint sum has degree of exactness 1 (Exercise 1.1).



(a) Left sum L_4. **(b)** Right sum R_4. **(c)** Midpoint sum M_4.

Figure 1.1: Three Riemann sums for e^{-x^2} on $[0, 1]$ with $n = 4$ cells.

1.3 Measuring convergence

Let $E_n = |I - Q_n|$ be the error of a rule using n cells of width $h = (b-a)/n$. We say that the rule has *order* p if $E_n \approx Ch^p$ for some constant C as $h \to 0$. If we double n, the error is divided by about 2^p, so that

$$p \approx \log_2 \frac{E_n}{E_{2n}}. \tag{1.2}$$

Table 1.1 gives the errors of the three Riemann sums on the running example. From $n = 16$ to $n = 32$, the left sum improves by a factor $1.95/0.982 \approx 1.99$, hence $p \approx 1$; the midpoint sum improves by a factor $1.20 \cdot 10^{-4}/2.99 \cdot 10^{-5} \approx 4.0$, hence $p \approx 2$.

Table 1.1: Absolute errors of the Riemann sums for $\int_0^1 e^{-x^2}\,\mathrm{d}x$.

n	$\lvert I - L_n\rvert$	$\lvert I - R_n\rvert$	$\lvert I - M_n\rvert$
4	7.52×10^{-2}	8.29×10^{-2}	1.92×10^{-3}
8	3.85×10^{-2}	4.05×10^{-2}	4.79×10^{-4}
16	1.95×10^{-2}	2.00×10^{-2}	1.20×10^{-4}
32	9.82×10^{-3}	9.94×10^{-3}	2.99×10^{-5}

Order alone does not tell the whole story, because the constant C also matters; but it describes how fast a method becomes accurate when the work is increased. The goal of the next chapters is to find rules of much higher order.

Exercises

Exercise 1.1. Show that the one-cell midpoint rule $M(f) = (b-a)f\left(\frac{a+b}{2}\right)$ has degree of exactness exactly 1.

Exercise 1.2. Use Table 1.1 and formula (1.2) to estimate the order of the right sum R_n from $n = 8$ to $n = 16$ and from $n = 16$ to $n = 32$.

Exercise 1.3. Show that if f is increasing on $[a, b]$, then $L_n \leq I \leq R_n$. What changes for the decreasing function e^{-x^2}?

2

Newton–Cotes Rules

2.1 Interpolatory quadrature

A natural way to build a rule is to replace f by a polynomial that is easy to integrate. Given nodes $x_0, \dots, x_n$, let p_n be the polynomial of degree at most n that interpolates f at these nodes. In the Lagrange form, $p_n(x) = \sum_{i=0}^{n} f(x_i)\,\ell_i(x)$ with $\ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$, so that

$$Q(f) = \int_a^b p_n(x)\,\mathrm{d}x = \sum_{i=0}^{n} w_i f(x_i), \qquad w_i = \int_a^b \ell_i(x)\,\mathrm{d}x.$$

Theorem 2.1: Degree of exactness of interpolatory rules

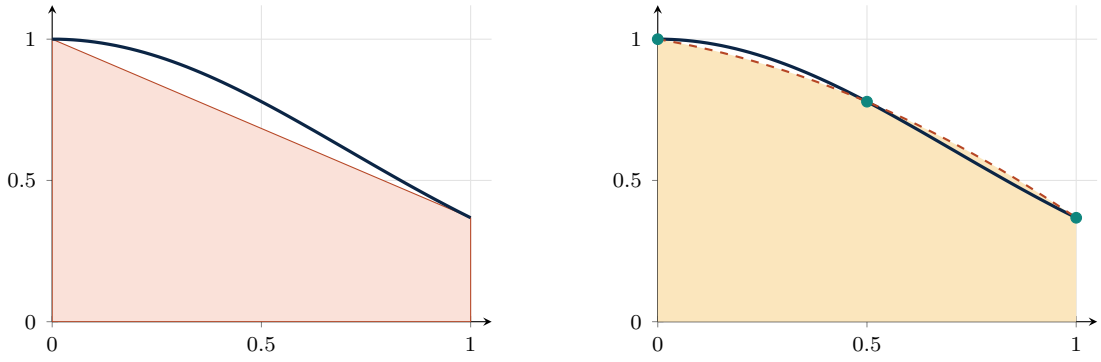
An interpolatory rule with $n+1$ nodes has degree of exactness at least n.

Proof. If f is a polynomial of degree at most n, then it coincides with its interpolation polynomial, $p_n = f$, so $Q(f) = \int_a^b p_n = \int_a^b f$. □

When the nodes are equally spaced, the rules are called *Newton–Cotes rules*. With $n = 1$ one obtains the *trapezoidal rule*, and with $n = 2$ *Simpson's rule*. On an interval $[x_0, x_0 + 2h]$ with midpoint x_1 they read

$$\frac{h}{2}\big[f(x_0) + f(x_0 + h)\big] \text{ (one cell of width } h\text{)}, \qquad \frac{h}{3}\big[f(x_0) + 4f(x_1) + f(x_0 + 2h)\big].$$

The weights of Simpson's rule are derived in Exercise 2.1. Figure 2.1 compares the two ideas on the running example: a line through two points, a parabola through three.



(a) One trapezoid.

(b) One parabola through three points.

Figure 2.1: The trapezoidal rule and Simpson's rule on $[0, 1]$ with $f(x) = e^{-x^2}$.

Theorem 2.2: Simpson's rule is exact for cubics

Simpson's rule has degree of exactness 3, although it uses only three nodes.

Proof. By Theorem 2.1, the degree is at least 2. By an affine change of variable we may work on $[-1,1]$, with nodes $-1,0,1$ and weights $\tfrac{1}{3},\tfrac{4}{3},\tfrac{1}{3}$. For $f(x)=x^3$, the rule gives $\tfrac{1}{3}(-1)+\tfrac{4}{3}\cdot 0+\tfrac{1}{3}(1)=0=\int_{-1}^{1}x^3\,dx$, because both the rule and the integral are symmetric. Hence the rule is exact for $1,x,x^2,x^3$, and so for all cubics. It is not exact for x^4 (Exercise 2.4). □

2.2 Composite rules and error estimates

To reach high accuracy one does not increase the degree of the polynomial, which is numerically unstable for equally spaced nodes, but one subdivides the interval. With $h=(b-a)/n$ and $x_i=a+ih$, the *composite trapezoidal rule* and the *composite Simpson rule* (with n even) are

$$T_n = h\Big[\tfrac{1}{2}f(x_0)+\sum_{i=1}^{n-1}f(x_i)+\tfrac{1}{2}f(x_n)\Big], \tag{2.1}$$

$$S_n = \tfrac{h}{3}\Big[f(x_0)+4\sum_{i\ \text{odd}}f(x_i)+2\sum_{\substack{i\ \text{even}\\ 2\le i\le n-2}}f(x_i)+f(x_n)\Big]. \tag{2.2}$$

Theorem 2.3: Composite error estimates

If $f\in C^2[a,b]$, there is $\xi\in(a,b)$ such that $I-T_n=-\dfrac{(b-a)\,h^2}{12}f''(\xi)$. If $f\in C^4[a,b]$ and n is even, there is $\eta\in(a,b)$ such that $I-S_n=-\dfrac{(b-a)\,h^4}{180}f^{(4)}(\eta)$.

Thus the trapezoidal rule has order 2 and Simpson's rule has order 4.

Example 2.4: Checking the bounds

For $f(x)=e^{-x^2}$ one has $f''(x)=(4x^2-2)e^{-x^2}$ and $f^{(4)}(x)=(16x^4-48x^2+12)e^{-x^2}$, with $\max_{[0,1]}|f''|=2$ and $\max_{[0,1]}|f^{(4)}|=12$ (both attained at $x=0$). For $n=4$, i.e. $h=\tfrac{1}{4}$, Theorem 2.3 gives $|I-T_4|\le\frac{(1/4)^2}{12}\cdot 2\approx 1.0\cdot 10^{-2}$ and $|I-S_4|\le\frac{(1/4)^4}{180}\cdot 12\approx 2.6\cdot 10^{-4}$. The actual errors are $3.8\cdot 10^{-3}$ and $3.1\cdot 10^{-5}$: the bounds are correct but pessimistic.

The quantity $f''(\xi)$ is unknown, so the theorem is used to *predict the order* and to obtain guaranteed (if pessimistic) bounds, as in Exercise 2.3. Figure 2.2 confirms the orders on the running example: the two curves are straight lines in a log–log plot, with slopes -2 and -4.

2.3 Richardson extrapolation and Romberg's method

The trapezoidal error has a remarkable structure. By the Euler–Maclaurin formula, for smooth f,

$$T_n - I = \frac{h^2}{12}\big[f'(b)-f'(a)\big]-\frac{h^4}{720}\big[f'''(b)-f'''(a)\big]+O(h^6). \tag{2.3}$$

Since the leading term is $c\,h^2$ with c independent of h, a combination of T_n and T_{2n} eliminates it:

$$\frac{4T_{2n}-T_n}{3} = I+O(h^4). \tag{2.4}$$

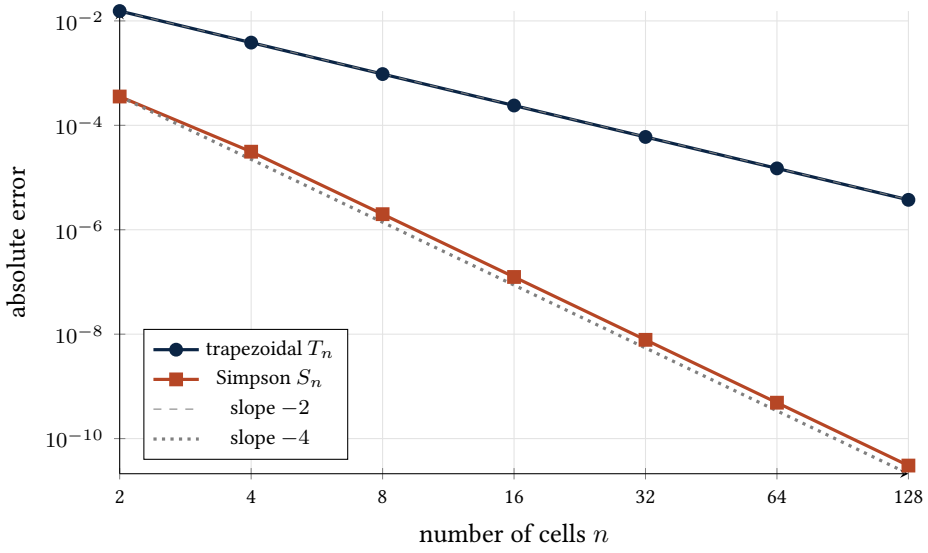


Figure 2.2: Errors of the composite trapezoidal and Simpson rules for $\int_0^1 e^{-x^2}\,\mathrm{d}x$.

A direct computation (Exercise 2.2) shows that the left-hand side is exactly Simpson's rule S_{2n}. Repeating the idea on the new h^4 term, and so on, yields *Romberg's method*:

$$R_{k,0} = T_{2^k}, \qquad R_{k,j} = \frac{4^j R_{k,j-1} - R_{k-1,j-1}}{4^j - 1} \quad (1 \le j \le k).$$

Table 2.1 shows the Romberg table for the running example: with only 16 cells, the extrapolated values agree with $I = 0.7468241328$ to nine digits.

Table 2.1: Romberg table for $\int_0^1 e^{-x^2}\,\mathrm{d}x$ ($n = 2^k$ cells; exact value 0.7468241328).

n	$R_{k,0}$	$R_{k,1}$	$R_{k,2}$	$R_{k,3}$	$R_{k,4}$
1	0.6839397206				
2	0.7313702518	0.7471804289			
4	0.7429840978	0.7468553798	0.7468337098		
8	0.7458656148	0.7468261205	0.7468241699	0.7468240185	
16	0.7465845968	0.7468242574	0.7468241332	0.7468241326	0.7468241331

Exercises

Exercise 2.1. Find the weights w_0, w_1, w_2 of the rule $\int_{-1}^{1} f \approx w_0 f(-1) + w_1 f(0) + w_2 f(1)$ by requiring exactness for 1, x and x^2.

Exercise 2.2. Let $h = (b-a)/(2n)$. Show that $\frac{4T_{2n} - T_n}{3} = S_{2n}$.

Exercise 2.3. Using $\max_{[0,1]} |f''| = 2$ for $f(x) = e^{-x^2}$, find the smallest n for which Theorem 2.3 guarantees $|I - T_n| \le 10^{-6}$.

Exercise 2.4. Apply Simpson's rule with one pair of cells on $[0, 1]$ to x^4. Compare with the exact integral and with the error formula of Theorem 2.3.

3

Gaussian Quadrature

3.1 Freeing the nodes

In a Newton–Cotes rule the nodes are imposed and only the weights are chosen. If we also choose the nodes, we have $2m$ parameters for an m-node rule, and we may hope for degree of exactness $2m-1$. The first nontrivial case shows that this works.

Example 3.1: The two-point rule

We look for nodes $\pm t$ and weights $w_1 = w_2 = w$ (by symmetry) such that $wf(-t) + wf(t) \approx \int_{-1}^{1} f$. Exactness for $f = 1$ gives $2w = 2$, so $w = 1$. Exactness for $f = x^2$ gives $2t^2 = \frac{2}{3}$, so $t = \pm\frac{1}{\sqrt{3}} \approx \pm 0.5773502692$. The rule is then automatically exact for x and x^3 (odd functions), hence for all cubics: two nodes give degree $3 = 2 \cdot 2 - 1$.

3.2 Legendre polynomials

The optimal nodes turn out to be the zeros of orthogonal polynomials. The *Legendre polynomials* P_n can be defined by Rodrigues' formula

$$P_n(x) = \frac{1}{2^n n!} \frac{\mathrm{d}^n}{\mathrm{d}x^n} \left(x^2 - 1\right)^n,$$

or by the recurrence $P_0 = 1$, $P_1 = x$ and $(n+1)P_{n+1}(x) = (2n+1)\,x\,P_n(x) - n\,P_{n-1}(x)$. The first ones are

$$P_2 = \tfrac{1}{2}(3x^2-1), \quad P_3 = \tfrac{1}{2}(5x^3-3x), \quad P_4 = \tfrac{1}{8}(35x^4-30x^2+3), \quad P_5 = \tfrac{1}{8}(63x^5-70x^3+15x),$$

and they are drawn in Figure 3.1. Their key property is *orthogonality*:

$$\int_{-1}^{1} P_m(x) P_n(x)\,\mathrm{d}x = \frac{2}{2n+1}\,\delta_{mn}. \tag{3.1}$$

In particular, P_m is orthogonal to every polynomial of degree less than m, and it has m distinct real zeros in $(-1, 1)$.

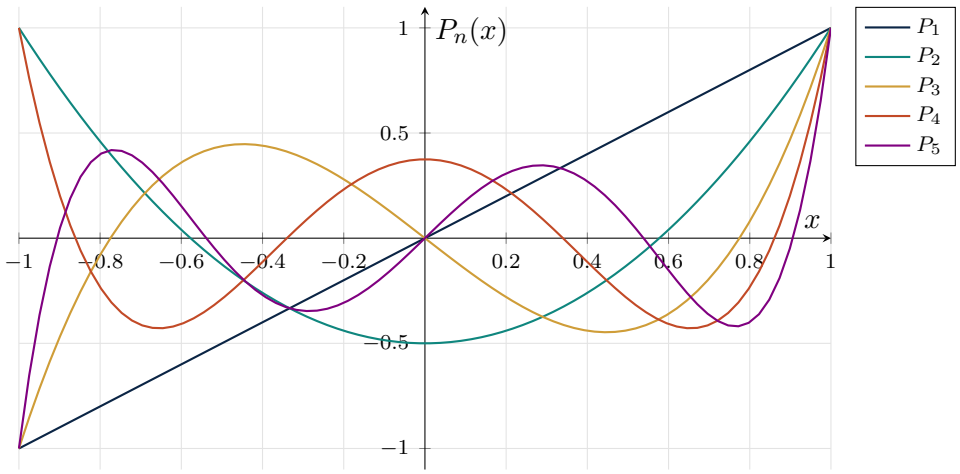


Figure 3.1: The Legendre polynomials $P_1, \dots, P_5$ on $[-1, 1]$.

3.3 The Gauss–Legendre rule

Theorem 3.2: Gauss–Legendre quadrature

Let $t_1, \dots, t_m$ be the zeros of P_m and let

$$w_i = \frac{2}{(1 - t_i^2)\,[P_m'(t_i)]^2}, \qquad i = 1, \dots, m.$$

Then the rule $G_m(f) = \sum_{i=1}^{m} w_i f(t_i)$ has degree of exactness $2m - 1$ on $[-1, 1]$, and no m-node rule has a higher one. If $f \in C^{2m}[-1, 1]$, then for some $\xi \in (-1, 1)$

$$\int_{-1}^{1} f(x)\,\mathrm{d}x - G_m(f) = \frac{2^{2m+1}(m!)^4}{(2m+1)\,[(2m)!]^3}\, f^{(2m)}(\xi).$$

Proof of the degree of exactness. Take a polynomial p of degree at most $2m - 1$ and divide it by P_m: $p = qP_m + r$ with $\deg q \le m - 1$ and $\deg r \le m - 1$. By orthogonality, $\int_{-1}^{1} qP_m = 0$, so $\int p = \int r$. Since $P_m(t_i) = 0$, we have $p(t_i) = r(t_i)$, hence $G_m(p) = G_m(r)$. The rule with the nodes t_i and the weights above is interpolatory, so it is exact for r by Theorem 2.1. Therefore $G_m(p) = G_m(r) = \int r = \int p$. □

On a general interval $[a, b]$, the change of variable $x = \frac{b-a}{2}t + \frac{b+a}{2}$ gives

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{2} \sum_{i=1}^{m} w_i\, f\!\left(\frac{b-a}{2}t_i + \frac{b+a}{2}\right).$$

The nodes and weights for small m are listed in Table 3.1 (the negative nodes are the mirror images of the positive ones, with equal weights).

3.4 Comparison on the running example

Table 3.2 gives the Gauss–Legendre approximations of I on $[0, 1]$. With $m = 5$ nodes (five evaluations of f) the error is $6.1 \cdot 10^{-9}$. To reach a comparable accuracy, the composite Simpson rule needs $n = 32$ cells, that is 33 evaluations, which gives an error of $7.8 \cdot 10^{-9}$. For this analytic integrand the error of the Gaussian rule decreases much faster than any fixed power of $1/m$. This behaviour

Table 3.1: Gauss–Legendre nodes and weights on $[-1,1]$, computed by the script `book_data.py`.

m	nonnegative nodes t_i	weights w_i
2	0.5773502692	1.0000000000
3	0.0000000000	0.8888888889
	0.7745966692	0.5555555556
4	0.3399810436	0.6521451549
	0.8611363116	0.3478548451
5	0.0000000000	0.5688888889
	0.5384693101	0.4786286705
	0.9061798459	0.2369268851

is typical of Gaussian rules and is explained by the theory of polynomial approximation (Trefethen 2008; Trefethen 2013).

Table 3.2: Gauss–Legendre approximations of $\int_0^1 e^{-x^2}\,\mathrm{d}x$ with m nodes.

m	G_m	$\lvert I - G_m\rvert$
1	0.7788007831	3.20×10^{-2}
2	0.7465946883	2.29×10^{-4}
3	0.7468145842	9.55×10^{-6}
4	0.7468244681	3.35×10^{-7}
5	0.7468241268	6.05×10^{-9}
6	0.7468241329	7.77×10^{-11}

3.5 Other weight functions

Orthogonality with respect to a different weight leads to other Gaussian rules. Gauss–Chebyshev rules approximate $\int_{-1}^{1} f(x)(1-x^2)^{-1/2}\,\mathrm{d}x$, Gauss–Laguerre rules approximate $\int_0^\infty f(x)e^{-x}\,\mathrm{d}x$, and Gauss–Hermite rules approximate $\int_{-\infty}^{\infty} f(x)e^{-x^2}\,\mathrm{d}x$. The nodes and weights can be computed as eigenvalues and eigenvector components of a symmetric tridiagonal matrix (Golub and Welsch 1969).

Exercises

Exercise 3.1. Apply the two-point Gauss rule of Example 3.1 to x^4 on $[-1,1]$ and compare with the exact integral.

Exercise 3.2. The three-point rule has nodes $0, \pm\sqrt{3/5}$ and weights $\tfrac{8}{9}, \tfrac{5}{9}, \tfrac{5}{9}$. Use it to compute $\int_{-1}^{1} x^4\,\mathrm{d}x$.

Exercise 3.3. Show that the change of variable $x = \tfrac{b-a}{2}t + \tfrac{b+a}{2}$ multiplies all weights by $\tfrac{b-a}{2}$.

Exercise 3.4. Compute G_2 for $\int_0^1 e^{-x^2}\,\mathrm{d}x$ and compare with Table 3.2.

4

Quadrature in Practice

4.1 Rounding errors

In exact arithmetic the error of a convergent rule tends to 0 as n grows. On a computer it does not: each evaluation of f and each addition is rounded, with relative error at most the *machine epsilon* $\varepsilon \approx 2.2 \cdot 10^{-16}$ in double precision. Quadrature rules with positive weights, like all those of this book, are numerically stable, because the rounding errors on the $f(x_i)$ are multiplied by weights that add up to $b - a$. In practice, the total error of a very fine composite rule is the sum of the *truncation error*, which decreases with h, and of rounding errors, which can slowly increase with n. This is why the error curves of Chapter 2 and of the Gaussian rules stop decreasing around 10^{-16}.

4.2 Adaptive Simpson quadrature

A composite rule uses the same step everywhere, although most of the error often comes from a small part of the interval, near a singularity or a rapid variation. *Adaptive quadrature* refines the mesh only where needed. For Simpson's rule the idea is the following. Let S be Simpson's rule on $[a, b]$ and let S' be the sum of Simpson's rules on the two halves. Since the error is proportional to h^4, halving h divides the error by 16, so $I - S' \approx \frac{1}{15}(S' - S)$, and therefore

$$I \approx S' + \frac{S' - S}{15}.$$

If $|S' - S| \le 15\varepsilon_{\text{tol}}$ the interval is accepted and this corrected value is returned; otherwise both halves are processed recursively with tolerance $\varepsilon_{\text{tol}}/2$ (Exercise 4.1). The box below summarises the method, and the listing after it gives a Python implementation.

Algorithm: adaptive Simpson quadrature

function Rec(f, a, b, ε, S)
  $m \leftarrow (a + b)/2$
  $S_L \leftarrow \frac{m-a}{6}\big[f(a) + 4f(\frac{a+m}{2}) + f(m)\big]$, $S_R \leftarrow \frac{b-m}{6}\big[f(m) + 4f(\frac{m+b}{2}) + f(b)\big]$
  if $|S_L + S_R - S| \le 15\varepsilon$ **then**
    return $S_L + S_R + (S_L + S_R - S)/15$
  else
    return Rec($f, a, m, \varepsilon/2, S_L$) + Rec($f, m, b, \varepsilon/2, S_R$)
  end if
end function

```
def adaptive_simpson(f, a, b, eps, maxdepth=50):
   def simpson(fa, fm, fb, a, b):
      return (b - a) / 6 * (fa + 4 * fm + fb)
```

```
    def rec(a, b, eps, whole, fa, fm, fb, depth):
        m = (a + b) / 2
        flm, frm = f((a + m) / 2), f((m + b) / 2)
        left = simpson(fa, flm, fm, a, m)
        right = simpson(fm, frm, fb, m, b)
        delta = left + right - whole
        if depth <= 0 or abs(delta) <= 15 * eps:
            return left + right + delta / 15
        return (rec(a, m, eps / 2, left, fa, flm, fm, depth - 1) +
                rec(m, b, eps / 2, right, fm, frm, fb, depth - 1))

    fa, fb, fm = f(a), f(b), f((a + b) / 2)
    return rec(a, b, eps, simpson(fa, fm, fb, a, b), fa, fm, fb, maxdepth)
```

Table 4.1 reports the results for three integrands and a requested tolerance of 10^{-8}. The method is conservative: the actual errors are about 10^{-11}. The integrand $\sqrt{x}$ needs many more evaluations than e^{-x^2}, because its derivative is infinite at 0. Figure 4.1 shows why: the algorithm automatically concentrates small subintervals near the singularity, where the width of the cells decreases proportionally to the distance to 0.

Table 4.1: Adaptive Simpson quadrature with tolerance 10^{-8}.

Integrand	evaluations	subintervals	error
e^{-x^2} on $[0, 1]$	61	15	1.10×10^{-11}
$\sqrt{x}$ on $[0, 1]$	317	79	2.55×10^{-11}
$1/(1 + 25x^2)$ on $[-1, 1]$	361	90	1.02×10^{-11}

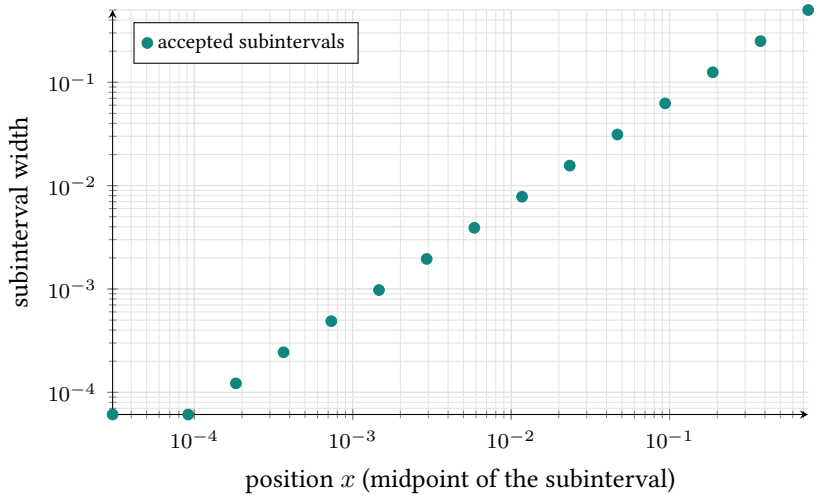


Figure 4.1: Final mesh of the adaptive Simpson rule for $\int_0^1 \sqrt{x}\,dx$ with tolerance 10^{-5}: the cells shrink toward the singularity at $x = 0$.

4.3 Choosing a method

No rule is best in all situations. The decision tree of Figure 4.2 summarises the advice of this book, and Table 4.2 lists the main properties of the rules studied.

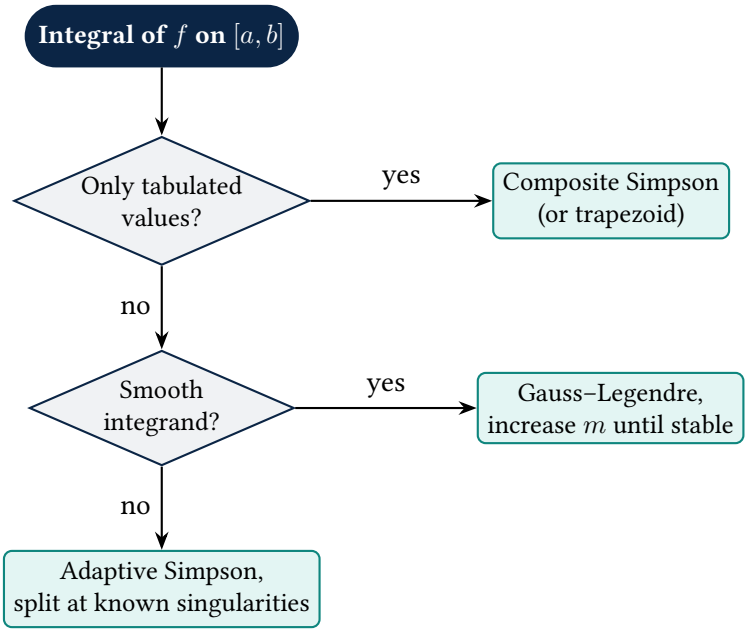


Figure 4.2: A simple decision tree for choosing a one-dimensional quadrature rule.

Table 4.2: Summary of the rules studied in this book.

Rule	Degree	Order	Typical use
Left/right sum	0	h	illustration only
Midpoint	1	h^2	open rules, simple codes
Trapezoidal	1	h^2	tabulated or periodic data
Simpson	3	h^4	robust default for smooth data
Romberg	increasing	h^{2k}	smooth integrands, reuse of T_n
Gauss–Legendre (m nodes)	$2m-1$	fast for analytic f	smooth integrands
Adaptive Simpson	local	tolerance driven	singularities, sharp features

For further reading, Davis and Rabinowitz (1984) is the classical reference on quadrature, Burden, Faires, and Burden (2016) and Press et al. (2007) give accessible introductions with code, and Trefethen (2013) develops the approximation theory that explains why Gaussian rules converge so quickly.

Exercises

Exercise 4.1. The error of Simpson's rule on an interval of length L is proportional to L^5 (for a fixed smooth f, this is the local form of the h^4 estimate). Use this to explain the constant 15 in $I \approx S' + (S' - S)/15$.

Exercise 4.2. Explain, using Figure 4.1, why Table 4.1 reports many more evaluations for $\sqrt{x}$ than for e^{-x^2}.

A

Notation

Table A.1: Symbols used in this book.

Symbol	Meaning
I	exact value of the integral $\int_a^b f(x)\,\mathrm{d}x$
h	cell width, $h = (b-a)/n$
x_i	equally spaced nodes, $x_i = a + ih$
L_n, R_n, M_n	left, right and midpoint Riemann sums
T_n, S_n	composite trapezoidal and Simpson rules
$R_{k,j}$	entries of the Romberg table
P_n	Legendre polynomial of degree n
t_i, w_i	Gauss–Legendre nodes and weights on $[-1, 1]$
G_m	m-node Gauss–Legendre rule
ε	machine epsilon, about $2.2 \cdot 10^{-16}$

B

Solutions to the Exercises

Exercise 1.1. By a translation we may take the interval $[-h, h]$. Then $M(f) = 2hf(0)$. For $f = 1$: $M = 2h = \int$. For $f = x$: $M = 0 = \int_{-h}^{h} x\,\mathrm{d}x$. For $f = x^2$: $M = 0$ while $\int_{-h}^{h} x^2\,\mathrm{d}x = \frac{2h^3}{3} \neq 0$. So the degree of exactness is exactly 1.

Exercise 1.2. From Table 1.1: $\log_2 \frac{4.05\cdot 10^{-2}}{2.00\cdot 10^{-2}} \approx 1.02$ and $\log_2 \frac{2.00\cdot 10^{-2}}{9.94\cdot 10^{-3}} \approx 1.01$. The order is 1.

Exercise 1.3. If f is increasing, then on each cell $[x_i, x_{i+1}]$ we have $f(x_i) \le f(x) \le f(x_{i+1})$, so $hf(x_i) \le \int_{x_i}^{x_{i+1}} f \le hf(x_{i+1})$. Summing over the cells gives $L_n \le I \le R_n$. For a decreasing function the inequalities are reversed: $R_n \le I \le L_n$. This agrees with the running example, where $R_4 = 0.6640 < I = 0.7468 < L_4 = 0.8220$.

Exercise 2.1. Exactness for $1, x, x^2$ gives $w_0 + w_1 + w_2 = 2$, $-w_0 + w_2 = 0$ and $w_0 + w_2 = \frac{2}{3}$. Hence $w_0 = w_2 = \frac{1}{3}$ and $w_1 = 2 - \frac{2}{3} = \frac{4}{3}$: Simpson's weights $\frac{1}{3}, \frac{4}{3}, \frac{1}{3}$.

Exercise 2.2. Write $f_i = f(a + ih)$ with $h = (b-a)/(2n)$. Then $T_{2n} = h\big[\frac{f_0}{2} + \sum_{i=1}^{2n-1} f_i + \frac{f_{2n}}{2}\big]$ and, with cells of width $2h$, $T_n = h\big[f_0 + 2\sum_{i \text{ even}} f_i + f_{2n}\big]$ (sum over even i with $2 \le i \le 2n-2$). Therefore $4T_{2n} - T_n = h\big[f_0 + 4\sum_{i \text{ odd}} f_i + 2\sum_{i \text{ even}} f_i + f_{2n}\big] = 3S_{2n}$. Numerically, $(4T_4 - T_2)/3 = (4 \cdot 0.7429840978 - 0.7313702518)/3 = 0.7468553798 = S_4$.

Exercise 2.3. We need $\frac{(b-a)h^2}{12} \cdot 2 \le 10^{-6}$ with $b - a = 1$, i.e. $h^2 \le 6 \cdot 10^{-6}$, $h \le 2.449 \cdot 10^{-3}$, so $n \ge 1/h = 408.2$. The smallest guaranteed n is 409. (The actual error is far smaller, since the bound is pessimistic.)

Exercise 2.4. With nodes $0, \frac{1}{2}, 1$: $S = \frac{1}{6}\big[0 + 4 \cdot \frac{1}{16} + 1\big] = \frac{5}{24} \approx 0.208333$, while $\int_0^1 x^4\,\mathrm{d}x = \frac{1}{5}$. The error is $I - S = \frac{1}{5} - \frac{5}{24} = -\frac{1}{120}$. The formula gives $-\frac{(b-a)h^4}{180} f^{(4)}(\eta) = -\frac{(1/2)^4}{180} \cdot 24 = -\frac{1}{120}$, since $f^{(4)} = 24$ is constant: the formula is exact here.

Exercise 3.1. The rule gives $1 \cdot \big(\frac{1}{\sqrt{3}}\big)^4 + 1 \cdot \big(\frac{1}{\sqrt{3}}\big)^4 = \frac{2}{9}$, while $\int_{-1}^{1} x^4\,\mathrm{d}x = \frac{2}{5}$. The rule is not exact for degree 4, as expected from a degree of exactness of 3.

Exercise 3.2. $\frac{5}{9}\big(\sqrt{3/5}\big)^4 \cdot 2 + \frac{8}{9} \cdot 0 = \frac{10}{9} \cdot \frac{9}{25} = \frac{2}{5}$, which is exactly $\int_{-1}^{1} x^4\,\mathrm{d}x$, because the degree of exactness is $2 \cdot 3 - 1 = 5 \ge 4$.

Exercise 3.3. With $x = \frac{b-a}{2}t + \frac{b+a}{2}$ we have $\mathrm{d}x = \frac{b-a}{2}\,\mathrm{d}t$, so $\int_a^b f(x)\,\mathrm{d}x = \frac{b-a}{2}\int_{-1}^{1} f\big(x(t)\big)\,\mathrm{d}t \approx \frac{b-a}{2}\sum_i w_i f\big(x(t_i)\big)$: every weight is multiplied by $\frac{b-a}{2}$.

Exercise 3.4. On $[0,1]$, $x_{1,2} = \frac{1}{2} \mp \frac{1}{2\sqrt{3}} \approx 0.2113,\ 0.7887$ and both weights equal $\frac{1}{2}$. Thus $G_2 = \frac{1}{2}\big(e^{-0.04466} + e^{-0.62201}\big) \approx 0.7466$, in agreement with Table 3.2 ($G_2 = 0.7465946883$).

Exercise 4.1. Let $E = I - S \approx cL^5 f^{(4)}$ be the error of the rule on the whole interval of length L. Each half has length $L/2$, hence an error about $c(L/2)^5 f^{(4)} = E/32$ (taking $f^{(4)}$ roughly constant), and the two halves together give $I - S' \approx 2 \cdot E/32 = E/16$. Then $S' - S = (I - S) - (I - S') \approx E - \frac{E}{16} = \frac{15}{16}E$, hence $I - S' \approx \frac{E}{16} = \frac{S' - S}{15}$.

Exercise 4.2. The derivatives of $\sqrt{x}$ blow up at 0, so Simpson's rule has no longer order 4 on the cells that touch 0. The algorithm must halve these cells many times (the points in Figure 4.1 lie on a line of slope 1 in log–log scale, i.e. the width is proportional to the distance to 0), while for e^{-x^2} a few uniform halvings suffice.

Bibliography

Burden, Richard L., J. Douglas Faires, and Annette M. Burden (2016). *Numerical Analysis*. 10th ed. Boston: Cengage Learning.

Davis, Philip J. and Philip Rabinowitz (1984). *Methods of Numerical Integration*. 2nd ed. Orlando: Academic Press.

Golub, Gene H. and John H. Welsch (1969). "Calculation of Gauss Quadrature Rules". In: *Mathematics of Computation* 23.106, pp. 221–230.

Press, William H., Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery (2007). *Numerical Recipes: The Art of Scientific Computing*. 3rd ed. Cambridge: Cambridge University Press.

Trefethen, Lloyd N. (2008). "Is Gauss Quadrature Better than Clenshaw–Curtis?" In: *SIAM Review* 50.1, pp. 67–87.

— (2013). *Approximation Theory and Approximation Practice*. Philadelphia: SIAM.

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